User Perceptions and Design Guidelines for Multi-Agent LLM Interfaces

Keshav Trikha
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
ktrikha2@illinois.edu

Zaid Takriti
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
ztakri2@illinois.edu

Shunchang Chen
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
sc176@illinois.edu

Wenxiao Li
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
wl65@illinois.edu

Ronak Chaudhuri
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
rchau5@illinois.edu

Jiarui Yao
Siebel School of Computing and Data Science
University of Illinois Urbana-Champaign
jiarui14@illinois.edu

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Abstract
The emergence of multi-agent systems leveraging Large Language Models (LLMs) has enabled specialized agents to collaborate on tasks such as retrieval, reasoning, and decision making. While technical research has focused on performance and protocols, less is known about user perceptions, trust, and coordination in these environments. We conducted a within-subject study (N=11), comparing four interface designs (Siloed, Sequential, Chat-based, Dashboard) for a two-agent meeting assistant. Results show that transparent coordination, clear role labeling, and structured process feedback significantly improve mental models, trust, and usability. We derive design guidelines to support effective human-agent collaboration in multi-agent LLM systems.

Author Keywords
Multi-Agent Systems; LLM Agents; Trust; Interface Design; Human-AI Collaboration

Introduction
In recent years, significant advancements in the capabilities of Large Language Models have extended their application beyond single-model interactions. A notable new technology is the emergence of multi-agent systems, where multiple LLM-based agents interact and collaborate within a shared interface. By distributing tasks such as retrieval, rea-

soning, and decision making, multi-agent architectures can lead to improvements in efficiency, accuracy, and adaptability compared to single-agent systems.

Despite rapid technical progress, most work to date has focused on system performance and coordination protocols, with comparatively little attention to how everyday users perceive, trust, and coordinate with these ensembles of agents. Studies of single-agent AI highlight that users often form incomplete or inaccurate mental models of AI behavior, struggle to calibrate trust, and experience cognitive overload when interacting with opaque systems. It is important for us to develop these human-centered insights in order to utilize multi-agent systems effectively.

In this paper, we investigate how users perceive, trust, and coordinate with multiple specialized LLM agents in a single interface, and the design implications for supporting human-agent collaboration. We analyze multi-agent systems through three lenses: user mental models, system functionality, and design implications.

Related Work

User Mental Models

Users generally perceive multi-agent systems (MAS) as inherently more capable compared to single-agent systems, primarily due to the specialization of tasks among agents. According to User Perspectives on Approaches to Multi-agent Conversational AI [2], users often attribute greater expertise to MAS because specialized agents focus deeply on defined task domains such as coding, summarization, or translation. This task-specific expertise is perceived as a significant strength, enhancing the overall perceived capability of the system. However, these advantages can be mitigated by usability challenges. Users frequently experience difficulties in clearly identifying the roles and responsibilities of individual agents when the interface does not explicitly communicate these distinctions. This ambiguity can result in cognitive overload, particularly when users must manage simultaneous or conflicting responses from multiple agents. Thus, while specialization offers a promise of enhanced functionality, the complexity introduced by managing interactions among multiple specialized agents can compromise user experience.

Trust dynamics in multi-agent LLM systems are complex and multi-dimensional, influenced by both individual agent performance and inter-agent coordination. The study Beyond Self-Talk: A Communication-Centric Survey of LLM-Based Multi-Agent Systems [1] emphasizes that transparency in inter-agent communications significantly influences trust. When users have visibility into agents' internal processes—such as observing debates or explanations of reasoning—it reinforces confidence in the system's internal logic and decision-making processes. Trustworthiness is also substantially affected by the design elements related to turn-taking, clear identification of agent roles, and the consistency of agent responses. Conversely, when agents provide contradictory responses without clear reasoning or fail to converge toward coherent outcomes, users rapidly lose trust. Therefore, the transparency and coherence of communication within multi-agent interactions are critical determinants in fostering sustained user trust and engagement.

Effective coordination between users and agents in multi-agent systems heavily relies on well-designed interface affordances. According to LLMs Working in Harmony [9], interfaces must include clear agent labeling, visual differentiation of agent roles, and adaptive timing of agent responses to support seamless user-agent interactions. The authors advocate for orchestration layers that synthesize agent

interactions, presenting users with consolidated outputs rather than complex, raw interactions. Such orchestration helps prevent cognitive overload by minimizing user exposure to the complexities of agent interactions. Coordination also extends to user command strategies. Users generally prefer delegating tasks at a high level (e.g., assigning broad responsibilities like "You handle the finance part") rather than engaging in detailed micromanagement, which can substantially increase cognitive burdens. Moreover, effective coordination demands robust task-switching support mechanisms. The survey Large Language Model based Multi-Agents: A Survey [4] highlights that persistent memory, context-awareness, and centralized oversight mechanisms—whether maintained by the system or by users—are vital for ensuring continuity and effective control over task delegation and completion.

Overall, users perceive specialized LLM agents in multi-agent systems as capable and beneficial but simultaneously recognize the potential for cognitive overload and confusion if the interaction design is not carefully managed. Trust in these systems is fundamentally strengthened through transparency, consistency, and clear inter-agent communication. Furthermore, effective coordination relies on explicit agent role identification, centralized orchestration, and interfaces that allow intuitive delegation and management of tasks. As multi-agent systems continue to grow in prevalence and complexity, addressing these human-centered design factors will be paramount in achieving successful and trustworthy human-AI collaborations.

System Functionality and Architecture

The "Plan-Then-Execute" paradigm represents a foundational methodology in multi-agent systems (MAS), characterized by step-wise planning followed by incremental execution [5]. This approach inherently provides structured opportunities for user intervention, allowing for review and modification of agent outputs. While research has indicated that heightened user engagement through intervention can enhance task accuracy and facilitate error correction, it concurrently increases cognitive load and may negatively impact user confidence [5]. Balancing this user involvement with system autonomy is critical for optimizing both performance and usability in MAS.

Architecturally, MAS typically operate along five core dimensions. These include the definition of actor types—whether agent-agent, human-agent, or hybrid configurations [11] specification of collaboration types, ranging from fully cooperative to competitive or coopetitive scenarios [11]; establishment of system structures such as peer-to-peer networks, hierarchical coordination via central agents, or distributed setups [11]; employment of diverse collaboration strategies, including role-based and model-based assignments [11]; and implementation of coordination protocols that utilize payload referencing or dynamic routing to streamline agent communication [10]. Collectively, these architectural elements have demonstrated considerable efficacy, with multi-agent collaborations reportedly increasing goal success rates by up to 70% in enterprise settings compared to single-agent configurations [10]. Nevertheless, effectively calibrating trust remains a challenge, particularly as users can easily misplace trust in seemingly plausible yet flawed agent outputs [5].

Communication protocols form another integral aspect of MAS, significantly influencing user perceptions of system transparency and coordination effectiveness. Communication architectures—categorized into decentralized, centralized, layered, or nested forms—are selected based on specific task demands and environmental constraints [8]. Furthermore, the distinction between blackboard-based

systems, which utilize a shared knowledge repository, and message-based systems, which depend on direct inter-agent communication, distinctly shapes user experiences and mental models [8].

Memory management in MAS plays a pivotal role in user perceptions of agent capability. Advanced implementations utilize layered memory systems, allowing agents to concurrently handle immediate task context and retain longer-term knowledge [4]. Such sophisticated memory architectures enhance agents' contextual awareness and output consistency, directly impacting user trust formation. Agent profiling and specialization further influence user mental models by delineating clear roles and interaction dynamics, enabling systems to tackle complex and dynamic tasks through specialized collaborations [4]. Transparency in these specializations critically affects user comprehension of system abilities and limitations.

Debate and deliberation mechanisms within MAS, where agents explicitly challenge or refine each other's conclusions, have been shown to strengthen overall decision-making robustness [4]. These observable deliberations notably enhance user perceptions of system reliability. Dynamic agent routing, an advanced technical mechanism that selectively bypasses orchestration to reduce latency without compromising quality, also impacts user perceptions of system efficiency and responsiveness [10]. Payload referencing systems, where agents build directly upon each other's outputs, can significantly improve coordination efficiency, enhancing performance by as much as 23% in complex coding tasks [10]. Strategically designed human-in-the-loop intervention points enable effective user-agent collaboration, significantly influencing both system performance and trust levels. Systems that provide explicit "user edit" functionalities during planning stages exhibit measurable improvements in task outcomes [5]. Moreover, sophisticated perception mechanisms that integrate transformer-based encoding of visual inputs into agent cognition allow for consistent multimodal responsiveness, further shaping user interaction quality [8].

Modern MAS architectures encompass diverse frameworks tailored to specific control strategies and use cases. Hierarchical designs employ global planners to delegate tasks to specialized agents, enhancing modular execution and role clarity. Conversely, dynamic or equi-level architectures adjust agent roles contextually, prioritizing flexibility and robustness [7]. These choices profoundly affect user interpretations of agent behavior, accountability, and adaptability under uncertainty.

Effective MAS coordination aligns agent objectives, distributes tasks efficiently, and synchronizes inter-agent actions. Mechanisms like task decomposition, real-time role reassignment, and goal propagation maintain coherence and accommodate human-agent interactions by modeling user intent and collaborative planning [12]. Optimal coordination involves selective state sharing, dynamic responsibility reallocation, and minimization of redundant communications, ensuring both task continuity and user clarity. Decision-making strategies in MAS range from consensus-building deliberations to reinforcement learning techniques such as actor-critic or value decomposition methods. Transparent decision-making processes that expose intermediate reasoning and predictions tend to foster user trust, while opaque or overly automated systems risk user disengagement and misplaced trust [6].

Lastly, MAS communication approaches—whether structured message-passing, symbolic protocol exchanges, or natural language interactions—directly affect user perceptions of clarity and coordination. Systems providing visibility

into inter-agent dialogue through debate transcripts or decision logs enhance user understanding and confidence [3]. Conversely, limited transparency in inter-agent communication may obscure intent and diminish user trust, underscoring the importance of clear, informative, and accessible interaction mechanisms.

Interface Design Guidelines

Based on literature, we initially extracted the following guidelines, which will be further tested in our prototype study:

Role Clarity
Label each agent's function explicitly (e.g., "Summarizer Agent") with consistent titles, color coding, and module headers.

Agent Status Visibility
Integrate real-time status indicators to reveal agent activity (e.g., "Processing...", "Complete").

Coordination Transparency
Visualize agent handoffs and intermediate data flows using arrows, message displays, or structured diagrams.

Abstraction over Raw Logs
Summarize complex interactions in plain language with optional "expand for details" toggles for advanced users.

Integrated Yet Distinct Interaction Channels
In chat UIs, prefix responses with agent names/icons and maintain consistent formatting.

Error Recovery and Control Points
Provide checkpoints for editing or rejecting agent outputs before proceeding.

Methods

We conducted a controlled, within-subject user study with N=11 participants using a Wizard-of-Oz simulation of a multi-agent meeting assistant. Participants reported varying levels of AI familiarity: a majority used AI assistants like ChatGPT daily (63.6%) or weekly (27.3%). Technical knowledge about AI systems also varied, with many identifying as 'Advanced' (54.5%) or 'Intermediate' (36.4%). The task involved using the assistant to process selected meeting notes to get a summary of action items and suggestions for scheduling a follow-up meeting.

We tested four interface variations (presented to each participant in a fixed order: A, B, C, then D):

- **Interface A (Siloed):** Presented outputs from the Summarizer and Scheduler agents in separate, static panels, processed simultaneously after a single user action. (See Figure 1)
- **Interface B (Sequential):** Showed agent outputs sequentially, with status indicators (e.g., "Processing...", "Complete") for each agent and an explicit textual hand-off message indicating data transfer from the Summarizer to the Scheduler. (See Figure 2)
- **Interface C (Chat-based):** Used an integrated dialogue format where system messages and agent outputs (attributed to "Summarizer Agent" or "Scheduler Agent") appeared as messages in a single chat window. (See Figure 3)
- **Interface D (Dashboard):** Featured a dashboard layout with separate panels for each agent, dynamic status indicators, a progress bar for overall task completion, a visual arrow indicating data flow, and a simplified coordination log. (See Figure 4)

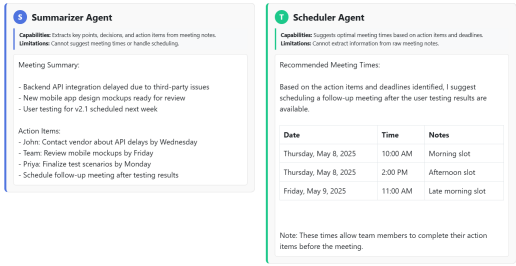

Figure 1: Interface A: Siloed design with separate panels for Summarizer and Scheduler outputs.

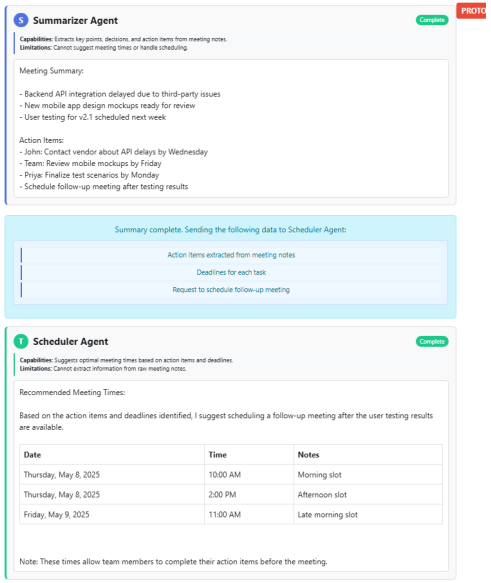

Figure 2: Interface B: Sequential design with status indicators and a textual hand-off message.

After interacting with each interface, participants completed

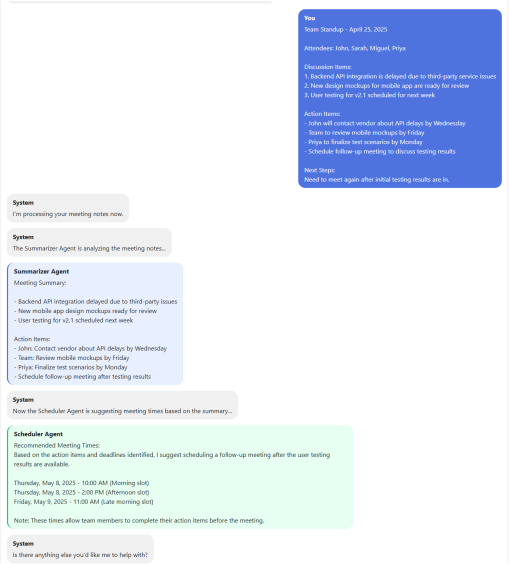

Figure 3: Interface C: Chat-based design with agent outputs attributed in a dialogue flow.

a post-task questionnaire. We collected quantitative ratings on 1-7 Likert scales for overall usability, understanding of agent actions, trust in the system, transparency of agent capabilities/limitations, visibility of coordination, and predictability of agent actions. Qualitative feedback was gathered through open-ended questions about their mental model of how the agents worked together, helpful or confusing aspects of the interface, and overall preferences.

Results

All 11 participants completed the study. Mean ratings (on a 1-7 scale, higher is better) for each interface across key metrics are summarized below.

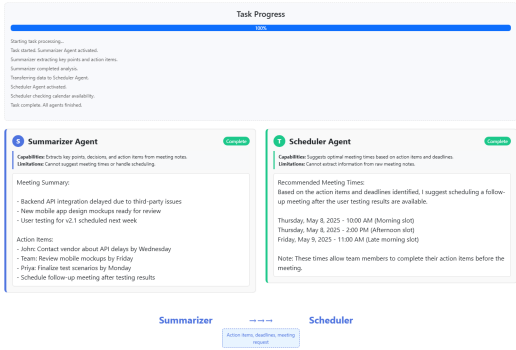

Figure 4: Interface D: Dashboard design with progress bar, visual data flow, and coordination log.

Quantitative Findings

Overall, Interface D (Dashboard) and Interface C (Chat-based) were rated higher across most metrics compared to Interface A (Siloed) and Interface B (Sequential).

- **Usability:** Interface C (M=6.27, SD=0.90), Interface D (M=6.09, SD=0.54), Interface B (M=6.09, SD=0.70), Interface A (M=5.55, SD=0.69).
- **Understanding of Agent Actions:** Interface C (M=6.55, SD=0.82), Interface D (M=6.73, SD=0.47), Interface B (M=6.45, SD=0.69), Interface A (M=5.91, SD=1.22).
- **Trust in System:** Interface C (M=6.09, SD=0.83), Interface D (M=6.09, SD=0.70), Interface B (M=6.00, SD=0.63), Interface A (M=5.00, SD=1.26).
- **Transparency of Capabilities/Limitations:** Interface D (M=6.27, SD=1.10), Interface B (M=6.09, SD=0.83), Interface C (M=5.64, SD=1.29), Interface A (M=5.09, SD=1.64). (Note: Interface C provided agent capabilities in an initial system message rather than persistently).
- **Coordination Visibility:** Interface C (M=6.64, SD=0.67), Interface D (M=6.55, SD=0.69), Interface B (M=6.36, SD=0.67), Interface A (M=4.64, SD=1.75).
- **Predictability of Agent Actions:** Interface C (M=6.36, SD=0.50), Interface D (M=6.09, SD=0.54), Interface B (M=6.09, SD=1.14), Interface A (M=5.36, SD=1.50).

Interface D was most often preferred overall (5 out of 11 participants), followed by Interface C (4 out of 11). Interface D also ranked highest for providing the clearest understanding of agent collaboration and being the most trusted system for the majority of users who specified a single interface. Transparency was also rated highly for Interface D.

Qualitative Themes

Qualitative feedback highlighted several key themes:

- **Clarity of Process and Coordination:** Participants highly valued interfaces that made the flow of information and the sequence of agent operations explicit. For Interface D, features like the progress bar, coordination log, and visual arrow indicating data transfer were frequently praised for enhancing understanding. For Interface B, the handoff message was seen as helpful in understanding the sequential nature. Interface C's conversational turn-taking and system messages describing agent actions also provided this clarity. In contrast, Interface A's simultaneous output was sometimes confusing regarding whether or how the agents coordinated.

- **Understanding Agent Roles and Specialization:** Clear labeling of agents and descriptions of their capabilities/limitations (present in all interfaces to some degree) were generally found helpful. However, when the coordination was unclear (as in Interface A), some users were less certain about how the distinct roles contributed to the final output.
- **Impact of Status Indicators:** Explicit status indicators (e.g., "Processing," "Complete") in Interfaces B and D were positively noted for providing feedback on the system's current activity.
- **Preference for Transparency:** Users expressed a preference for interfaces that showed "what the system was doing". The dashboard view (D), with its multiple cues, was often cited as the most transparent.
- **Familiarity of Chat (Interface C):** Some users found the chat interface (C) to be natural and easy to use due to its familiarity. However, others noted that for longer inputs/outputs, scrolling in a chat format could be less convenient than panel-based layouts for reviewing information. One user also experienced a formatting bug with the chat output.
- **Desire for Rationale and Control:** While not directly tested as a variable, some users expressed a desire for more insight into the "rationale" behind scheduler suggestions or the ability to intervene or customize outputs.
- **Perceived Need for Summarizer Refinement:** A recurring comment, irrespective of the interface, was that the Summarizer Agent's output often seemed too similar to the input notes, making its value less apparent. This feedback relates to the perceived agent capability rather than the interface design itself.

Recommendations and Design Implications

Based on our findings, we propose the following design implications and recommendations for creating more effective and trustworthy multi-agent LLM interfaces. These expand upon our initial guidelines and are directly informed by the study results:

1. **Prioritize Transparent Coordination Visualization:** Our study overwhelmingly showed that users benefit from explicit cues about inter-agent coordination. Interfaces like the Dashboard (D), which visualized data flow, showed task progression, and logged coordination steps, were rated highest for clarity and trust. Design Recommendation: *Visibly illustrate the flow of information and tasks between agents. Utilize progress indicators, summarized logs, and visual links to demystify the "black box" of inter-agent collaboration.* Avoid designs where agents appear to work in isolation without clear handoffs (as in Siloed A).
2. **Ensure Explicit Role Clarity and Attribution:** While labels are basic, clear differentiation is crucial. Users need to understand what each agent does and does not do. This was aided by the "Capabilities" and "Limitations" sections in our study. In dynamic interfaces like chat (C), persistent and clear attribution of messages to specific agents is vital for maintaining role understanding. Design Recommendation: *Clearly label each agent and its outputs. Consistently attribute*

actions and information to the responsible agent, especially in shared or conversational interfaces. Provide succinct descriptions of agent specializations.

3. **Provide Continuous Agent Status and Process Feedback:** Users appreciated knowing what agents were doing in real-time (e.g., "Processing," "Complete," "Sending data to Scheduler") as seen in Interfaces B, C, and D. This feedback helps manage expectations and reduces uncertainty. Design Recommendation: *Integrate dynamic status indicators for each agent and provide brief, human-readable updates on ongoing processes or state changes within the multi-agent workflow.*

4. **Balance Abstraction with Sufficient Detail:** While users do not need raw operational logs, a high-level overview of the coordination process, like the summarized log in Interface D, was found helpful. The desire for more "scheduling rationale" also indicates a need for more than just the final output. Design Recommendation: *Abstract complex internal agent communications, but provide summarized, understandable traces of key inter-agent actions and data transfers. Offer mechanisms for users to optionally access more detailed explanations or rationale if desired (Progressive Disclosure).*

5. **Leverage Familiar Interaction Patterns Carefully:** The Chat interface (C) benefited from user familiarity, leading to high usability. However, designers must consider the suitability of the paradigm for the task's information density. Design Recommendation: *Employ familiar interaction patterns (e.g., chat, dashboards) where appropriate, but adapt them to handle the specific needs of multi-agent interactions, such as clear agent attribution and managing potentially complex outputs from multiple sources.*

6. **Facilitate User Oversight and Perceived Control:** The qualitative feedback hinting at a desire for more insight into decision rationale and the ability to potentially intervene or customize outputs suggests the importance of user agency. Design Recommendation: *Design systems to not only be transparent but also to offer users opportunities for oversight. Consider points for user intervention, review, or customization of agent plans or outputs, particularly for critical tasks, to enhance trust and user satisfaction.*

Conclusion

This study investigated user perceptions of multi-agent LLM interfaces, highlighting the critical role of interface design in shaping mental models, trust, and usability. Our findings from a user study (N=11) comparing four distinct interface designs—Siloed, Sequential, Chat-based, and Dashboard—demonstrate that increased transparency in agent coordination and clear communication of agent roles significantly enhance user experience. Interfaces providing explicit feedback on agent status, data handoffs, and overall task progression, notably the Dashboard and Chat-based designs, were perceived more positively by users.

The derived design guidelines emphasize the importance of visualizing coordination, ensuring role clarity through explicit labeling and attribution, providing continuous status feedback, balancing abstraction with necessary detail, leveraging familiar interaction patterns thoughtfully, and facilitating user oversight. These principles offer actionable insights for designers aiming to create multi-agent systems that are not only technologically advanced but also intuitive and trustworthy. As multi-agent LLM systems become

more prevalent, adopting such human-centered design approaches will be paramount to fostering effective human-agent collaboration and realizing the full potential of these complex AI systems. Future work should explore these design principles across a wider range of tasks, user populations, and with fully autonomous agent behaviors to further validate and refine these guidelines.

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